

EE 3TR4 (COMMUNICATION SYSTEMS): QUIZ 3

INSTRUCTOR: DR. JUN CHEN

Student Name: _____

Student Number: _____

We wish to transmit a digital signal with a bit rate of 7Mb/s over a channel with a bandwidth of 5MHz. Should we use BPSK or QPSK? Justify your choice.

Next, provide a block diagram of a digital communication system (including transmitter, channel, and receiver) capable of transmitting a 7Mb/s signal over a 5MHz channel.

Describe the line coding scheme and explain how the bitstream is mapped to symbols. Finally, draw the corresponding constellation diagram.

Fourier transform of special functions:

$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f),$$

$$\text{triang}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}^2(f),$$

where

$$\text{rect}(t) = \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{triang}(t) = \begin{cases} 1 - |t|, & |t| < 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}.$$

Properties of Fourier transform:

If $g(t) \xleftrightarrow{\mathcal{F}} G(f)$ and $h(t) \xleftrightarrow{\mathcal{F}} H(f)$, then

$$g(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} G\left(\frac{f}{a}\right),$$

$$G(t) \xleftrightarrow{\mathcal{F}} g(-f),$$

$$g(t - t_0) \xleftrightarrow{\mathcal{F}} G(f) e^{-j2\pi f t_0},$$

$$g(t) e^{j2\pi f_c t} \xleftrightarrow{\mathcal{F}} G(f - f_c),$$

$$g(t) \cos(2\pi f_c t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [G(f - f_c) + G(f + f_c)],$$

$$g(t) * h(t) \xleftrightarrow{\mathcal{F}} G(f) H(f),$$

$$g(t) h(t) \xleftrightarrow{\mathcal{F}} G(f) * H(f).$$

Random process:

The autocorrelation function and power spectral density (PSD) form a Fourier transform pair:

$$R_X(\tau) \xleftrightarrow{\mathcal{F}} S_X(f).$$

For a linear time-invariant filter with transfer function $H(f)$, the input PSD $S_X(f)$ and output PSD $S_Y(f)$ are related by

$$S_Y(f) = S_X(f) |H(f)|^2.$$

Solution to Quiz 3

Since $\frac{7}{2} < 5 < 7$, QPSK satisfies the channel's 5MHz bandwidth limitation while BPSK does not.

Please refer to the lecture notes for the system block diagram, the line coding scheme, and the corresponding constellation diagram.

EE 3TR4 C02 (COMMUNICATION SYSTEMS): QUIZ 3

INSTRUCTOR: DR. TELEX M. N. NGATCHED

Student Name: _____

Student Number: _____

- 1) Explain what happens to a line-coded data sequence when passed through a severely limited bandlimited channel.
- 2) Draw the Unipolar NRZ, Polar NRZ, Unipolar RZ, and Bipolar RZ for the data sequence **1101001**.
- 3) Draw the block diagram of a digital communication system with DSB-SC (including transmitter, channel, and receiver).

1) line coded data seq. thru very limited bandlimited channel

↳ ISI

• high freq components attenuated

↳ causes sharp edge pulses to smear & spread in time

↳ causes temporal broadening to have individual symbols to overlap

↳ ISI

Fourier transform of special functions:

$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f),$$

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where

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Random process:

The autocorrelation function and power spectral density (PSD) form a Fourier transform pair:

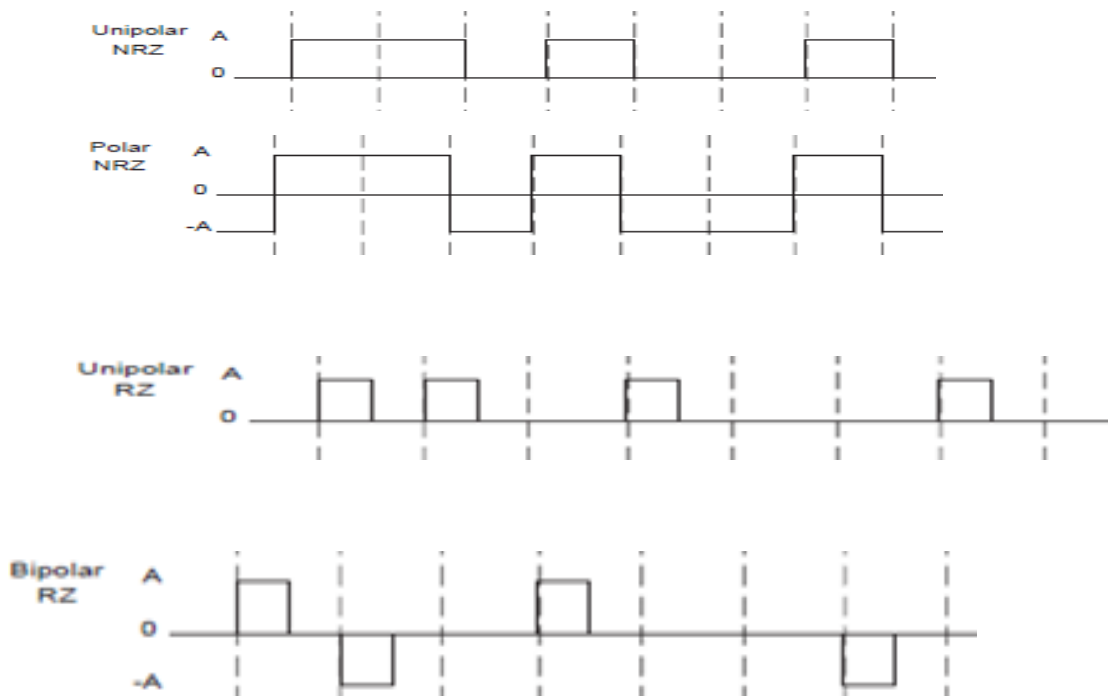
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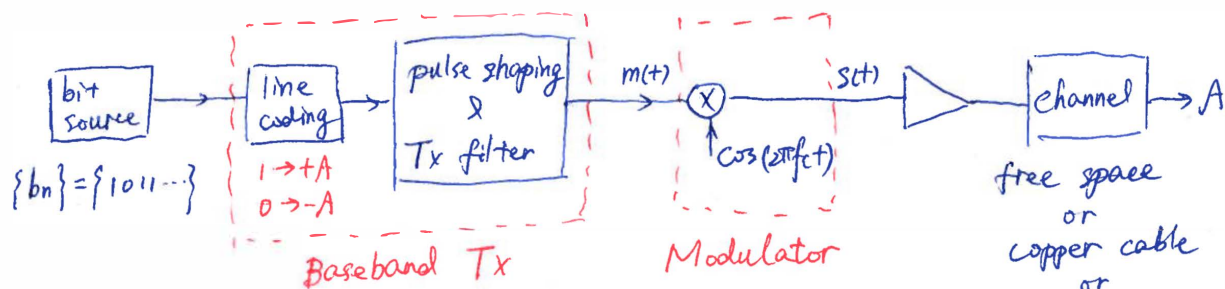
$$S_Y(f) = S_X(f) |H(f)|^2.$$

Solution to Quiz 3

- 1) When a line-coded data sequence is passed through a severely bandlimited channel, the high-frequency components of the signal are attenuated, causing the sharp-edged pulses to smear and spread in time. This temporal broadening causes individual symbols to overlap, leading to a phenomenon known as Inter-Symbol Interference (ISI). [2 Marks].
- 2) Line codes for the data sequence **1101001** [2x4 = 8 Marks].



- 3) Transmitter of a digital communication system with DSB-SC [5 Marks]..



Receiver of a digital communication system with DSB-SC [5 Marks]..

